

Chapter 5

NMR logging applications

There are numerous applications of NMR logging in formation evaluation for the estimation of hydrocarbon reserves and their producibility. As the logging tools improve their capability, more and more new applications of NMR logs are being discovered. The most frequently used applications now are: (1) porosity estimates, (2) irreducible water saturation determination, (3) permeability prediction, (4) residual oil estimation, (5) hydrocarbon typing, and (6) oil viscosity estimation. At the end of this section, we shall also discuss the logging guidelines, namely, some of the issues one should be aware of in the planning and preparation of an NMR logging run.

5.1 Porosity estimation

An NMR logging tool measures the total population of hydrogen nuclei in the formation, including the hydrogen atoms of the water and oil in the pore space, and those in the clay. Hydrogen atoms in the solid matrix, if any, are generally not sensed by the logging tools, due to the extremely long T_1 and short T_2 . The current NMR logging tools typically measure hydrogen atoms with relaxation times down to 1 ms, or in some cases, can be as low as 0.3 ms if the early signals are stacked many times. In general, the measured NMR signal only reflects the hydrogen nuclei in the pore space.

Because the measurements are sensitive to proton (i.e., hydrogen nucleus) spins in fluids only¹, the estimated porosity value should be mineralogy in-

¹There are exceptions. Sometimes, the logging conditions are such that the ^{23}Na signal is also measured [154] inadvertently.

dependent. As the crude oils have a hydrogen index very similar to that of water, porosity can be reliably measured in both water and oil productive zones. This measurement is considered *mineralogy independent* in reference to other porosity logging tools which have a significant response to the mineralogy as well as to pore fluids. The sensitivity of these other logging tools to mineralogy means that additional information or assumptions must be made to obtain porosity. However, there are several factors which could affect the NMR porosity estimates as well. These are discussed in the following.

Ferro- and paramagnetic materials

The presence of some iron bearing minerals or high magnetic susceptibility pore-lining materials can greatly reduce the T_2 relaxation times. As a result, some NMR signals with very short T_2 relaxation times are lost due to finite dead time of the instruments. This will lead to reduced porosity values. This situation is often accompanied by a shift of the T_2 peaks toward shorter relaxation times due to the large internal magnetic field gradient induced by the presence of these minerals. Thus, the NMR porosity determination is mineralogy independent, except when highly paramagnetic minerals are present.

Tool calibration

Improper tool calibration can lead to erroneous porosity readings. The measured NMR signal is calibrated against a known standard (typically a water sample) to give values in porosity units. The initial signal amplitude of a T_2 CPMG measurement (i.e., extrapolating the echo train back to zero time) is proportional to the density of hydrogen nuclei in the probed zone. Since this quantity depends on a number of factors, such as the rate of signal decay, the noise level of the measurement, the loading on the rf coil, etc., care should be taken to ensure it is properly done. Errors arising from tool calibration mistakes can often be easily identified because they affect the porosity value of the whole log.

Environmental corrections

There are a number of environmental corrections that need to be made before the measured NMR signal can yield the correct porosity values. First of all, the NMR amplitudes should be corrected for the temperature and field

strength to reflect their proportionality to the hydrogen spin density. From Curie's law:

$$M = N\gamma^2\hbar^2 I(I+1)H_0/3kT \quad (5.1)$$

where

$$\begin{aligned} M &= \text{magnetization} \\ N &= \text{spin density} \\ \gamma &= \text{gyromagnetic ratio} \\ \hbar &= \text{Planck constant}/2\pi \\ I &= \text{spin quantum number} \\ H_0 &= \text{magnetic field} \\ k &= \text{Boltzmann's constant} \\ T &= \text{absolute temperature} \end{aligned}$$

Thus, the magnetization, M , which is proportional to the NMR amplitude, increases with field strength, and decreases with increasing temperature, for the same amount of water volume. Hence, it must be corrected by a factor T/H_0 , i.e., the formation temperature and the logging frequency. The temperature and salinity also affect the hydrogen density of water. In addition to this, the salinity, thus the conductivity of the water, can affect the Q of the rf coil and reduce the signal strength. All of these are considered in the logging tool design and the signal value conversion, and are normally properly corrected by the logging companies. If some of these environmental corrections are inadvertently left out, or if special circumstances lead to improper corrections, the measured porosity values will be affected.

Insufficient polarization time

Under certain logging conditions (e.g., high temperature/long T_1 of pore fluids, vuggy carbonates, low surface relaxivities, ...), repeat delay times greater than 10 seconds are often required. Insufficient wait time between successive excitation pulses can cause incomplete recovery of the spins to their equilibrium state. This will lead to a reduction of the measured signal, and thus of the porosity, by the following factor

$$a = 1 - e^{-RD/T_1} \quad (5.2)$$

where RD stands for the Repeat Delay time (or wait time), i.e., the time between the end of a CPMG echo train and the start of the next CPMG excitation, and T_1 is the spin-lattice relaxation time for the fluid. This factor, α , is commonly known as the polarization factor. Typically, the wait time is chosen to be three times T_1 . This ensures that 95% of the spins have been polarized along the applied magnetic field direction.

The loss of porosity due to incomplete recovery can often be corrected by assuming a T_1/T_2 ratio and including the polarization factor in the data inversion process. Since, the T_1/T_2 ratio is not always a constant for the formation fluids for the whole logged depth, this exercise is sometimes not successful. This is especially true when there are multiple formation fluids such as water and oil, which can have very different T_1/T_2 ratios.

Presence of gas

The presence of free gas or gas dissolved in oil can also lead to errors in porosity. The hydrogen index (HI = the ratio of the signal from a volume of gas or oil to the signal from an equal volume of water) is significantly less than 1 even for very high pressure gas. Unless accounted for, this will cause the porosity to be underestimated.

Gas dissolved in oil reduces the viscosity of the oil without significantly increasing the oil volume. The viscosity reduction increases both T_2 and T_1 . These increases mean that longer wait times are needed with the potential for error during tool setup if the wait time is determined in a wet zone. The presence of gas in the oil may also decrease the HI of the oil which would lead to underestimation of porosity (see discussion in Chapter 3).

Presence of clay, heavy oils, or bitumen

When considering the contributions of clay to porosity, we touch upon an area of controversy. Some argue that the water of hydration in clays should be included as a part of the porosity, while others argue against it. Without taking sides in this controversy, we shall simply state that NMR measures the formation volume containing hydrogen nuclei with T_2 relaxation times greater than 1~3 ms, depending on the noise level. If a higher precision of measurement is achieved, the lower limit on T_2 relaxation time can be extended to 0.3 ms or less.

As discussed previously [177], current literature suggests that the T_2 of

clays covers a wide range of relaxation times from less than 0.1 to over 16 ms. Thus, in high precision measurements, most if not all of the porosity associated with clay will be included.

The T_2 relaxation times for heavy oils vary over a wide range depending on their viscosity. For viscous high density oils, the NMR log cannot capture all the signal and may suffer an apparent loss of porosity.

The T_2 relaxation time for solid bitumen is very short, often around 0.1 ms or less. Thus, solid bitumen cannot be measured with the current NMR logging tools. Because the neutron log measures bitumen and brine-saturated pore space, the combined information of neutron and NMR porosities may allow us to estimate the bitumen content of the formation.

Other factors

Additional sources of error in porosity are tool related. High well bore temperatures can cause tool malfunctioning. Cross talk between other wireline tools mounted on the same string can cause an erratic signal or an unacceptable noise level.

5.2 Irreducible water saturation

One of the important applications of NMR logging is estimating the irreducible water saturation. This is the portion of the capillary-bound water plus the clay-bound water in the pore space. Under a certain capillary pressure, this part of the water in the formation is not producible, hence the term, irreducible water. The knowledge of this information has at least two impacts: (1) determining the water cut, i.e., the amount of water to be produced from the reservoir, and (2) estimating the permeability by using the Coates equation as discussed in Chapter 4.

The irreducible water volume can be determined from NMR measurements/logs as discussed in Chapter 4. For using a T_2 cutoff to determine the irreducible water saturation, a T_2 relaxation time of 33 ms is typically chosen for sandstones, and 100 ms is used for carbonates. As discussed previously, these are values obtained based on a statistical average from experimental measurements. Different reservoirs may have very different rock fabrics with different capillary pressure behaviors. It is recommended that NMR core

measurements be performed for the reservoir so that a more specific T_2 cutoff, or a spectral method [64] can be applied.

To determine the water cut, or the producible water, one can subtract the irreducible water volume from the water volume estimated from the resistivity log. If the water saturation determined from the resistivity log, S_w , is significantly greater than the irreducible water saturation estimated from the NMR log, S_{wirr} , the reservoir will produce water early in its productive life. If $S_w \cong S_{wirr}$, the reservoir is at irreducible water saturation and no water will be produced in its productive life. In principle, the formation cannot have a water volume which is less than the irreducible water measured by NMR. However, such a situation is possible if the capillary pressure of the reservoir is higher than the air/water capillary pressure used in the laboratory for determining the NMR T_2 cutoff for irreducible water saturation.

The T_2 cutoff for sandstones appears to work best in well consolidated formations with permeabilities in the 0.3 md - 3,000 md range where the wells are drilled with water-based mud.

For poorly consolidated formations, reservoirs with long oil columns, formations with significant amounts of ferro- and paramagnetic compounds (e.g., iron, manganese), and reservoirs with heavy oil or bitumen, specific T_2 cutoffs may have to be established by core measurements.

Poorly consolidated clastics often have very low irreducible brine saturations in the range of 5-20%. For these rocks, the amount of T_2 signal below 33 ms is small and can be hard to resolve from the noise. Reservoirs with long oil columns may require cutoffs lower than 33 ms. Significant amounts of paramagnetic compounds on the grain surfaces can lead to large induced magnetic field gradients which shift the T_2 distributions to short times. Reservoirs with heavy oil can have T_2 distributions with overlapping oil and brine signals such that the irreducible water saturation, S_{wirr} , can not be readily resolved. Partially oil-wet reservoirs can also have overlapping T_2 distributions for the oil and brine.

The presence of heavy oil or bitumen, which has very short T_2 relaxation times, further complicates the issue. When one interprets the NMR T_2 distributions, one has to keep all these in mind.

Sources of T_2 values shorter than 3 ms include: water in extremely fine grain silts or clays, very heavy oils or bitumen, paramagnetic compounds in solution or on the grain surfaces, and hydrated water in gypsum or clay minerals.

5.3 Permeability prediction

As discussed in Chapter 4, there are two commonly used permeability estimators, i.e., the Coates/Timur equation [63, 214] and the $T_{2,LM}$ equation [121, 11]. Sezginer et al. [199] recently suggested a method that uses the summation of echoes to predict the permeability with some success.

Coates/Timur equation The Coates equation uses porosity, BVI, and FFI to predict permeability. Therefore, whenever the BVI and porosity can be accurately determined, this is the preferred method. However, the predicted value is quite sensitive to the value of BVI, and it generally tends to overestimate permeability.

$T_{2,LM}$ equation The $T_{2,LM}$ equation uses the logarithmic mean of T_2 distribution to predict permeability. The estimator is derived based on brine-saturated sedimentary rock samples. It generally gives good prediction for the water zone. However, in the oil zone, since the T_2 relaxation times for oils are in general different from those for water, the $T_{2,LM}$ equation is not reliable.

Summation of echoes The summation of echoes is related to the mean relaxation time, $\langle T_2 \rangle$, hence the surface to volume ratio of the large pores of the rock formation, and hence the permeability. Although laboratory experiments were carried out on a few quarried sandstone samples to show such a correlation [199], not enough laboratory measurements were performed to establish a more general and robust formula. It can be used as a qualitative indicator for permeability. For a more quantitative estimation, it is recommended that field specific formulas be established through NMR core measurements. One important advantage of this method is that it eliminates the noise by summing echoes in the time domain, thus retaining the high resolution of NMR information from pad-type tools such as Schlumberger's CMR without the need to stack the signals depth-wise.

With these three types of permeability estimators, we should be selective in properly applying them. As different reservoir conditions dictate different considerations in the application of these permeability estimators, we shall discuss a few as follows:

Water based mud, highly invaded reservoir

When water-based drilling muds are used and the formation is extensively invaded by mud filtrate, the $T_{2,LM}$ based method may be preferred because the $T_{2,LM}$ estimate is less sensitive to noise than is the BVI determination. This assumes that the reservoir is water-wet, and that all of the hydrocarbon has been flushed from near the well bore.

Water based mud, significant remaining low viscosity oil

When significant amounts of oil remain in the formation, the T_2 distribution may look bimodal, with the long time associated with oil where the oil has low viscosity and the reservoir is water-wet. Here, the Coates equation, or the $T_{2,LM}$ equation modified to account for the presence of oil, may be preferred.

Water based mud, significant remaining high viscosity oil

Neither equation will work as written when the oil and brine T_2 distributions overlap to the extent that they cannot be distinguished. Here, the oil and brine signals overlap, and the oil, though non-wetting, has a broad distribution of T_2 values. This is a difficult situation. No straightforward satisfactory solution has been found. One of the possible solutions is to combine diffusion measurements (discussed below), resistivity log and laboratory measurements of oil to determine the irreducible water saturation.

To illustrate the overlap between the oil and brine signals, the T_2 distributions for a reservoir rock sample are shown in Figure 5.1 before and after soaking in D_2O . The first measurement has both signals from the oil and brine whereas the second measurement has only the oil signal.

Oil based mud, low viscosity invading filtrate

When a well is drilled with oil based mud containing low viscosity oil filtrate, the Coates equation is preferred especially where the invading filtrate displaces the reservoir hydrocarbon. An example of this case is when the light oil filtrate displaces gas and results in a bimodal T_2 distribution where the brine and oil peaks are clearly separated. Irreducible brine saturation is determined directly from the area under the short time peak, and accurate estimates of permeability are obtained. Stonard et al. [206] used this approach to obtain the transform shown in Figure 5.2. Note that the transform

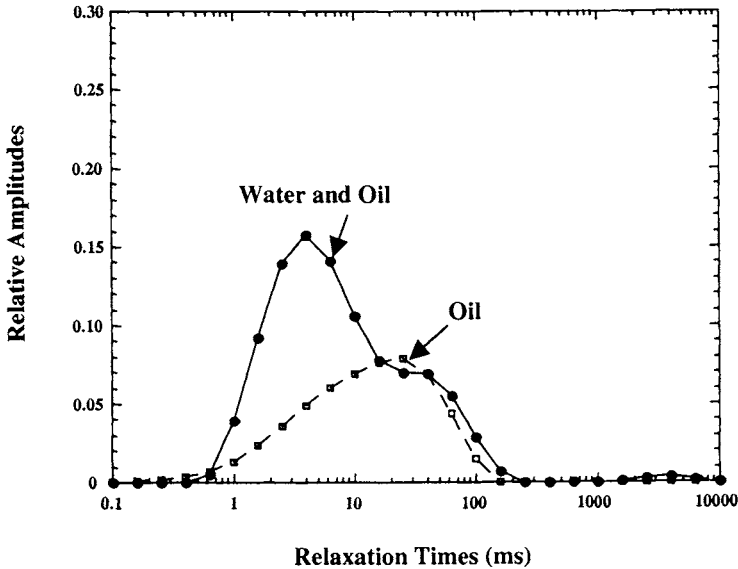


Figure 5.1: The T_2 distribution of a rock sample before and after soaking in D_2O to remove the water signal.

compares favorably with the core permeabilities through almost all of the log interval.

Field specific permeability algorithms

Even when the correct algorithm is chosen for a given field, the estimated permeabilities may be way off because of local conditions. The permeability algorithms supplied by the logging companies were based on laboratory measurements on a finite range of clastic reservoirs where, for example, the permeabilities were typically greater than 20 md. The laboratory based transform for the permeabilities of core plugs from various reservoirs can be very different from the commonly used transform $k = \phi^4 T_{2,LM}^2$.

Carbonate reservoirs

For carbonate reservoirs, it has long been known that NMR log derived permeability estimates are not as satisfactory and consistent as those for clastic sediments. One of the reasons for such difficulty is that there are intergranular and vuggy porosities. Chang et al. [56] found that if the porosity above

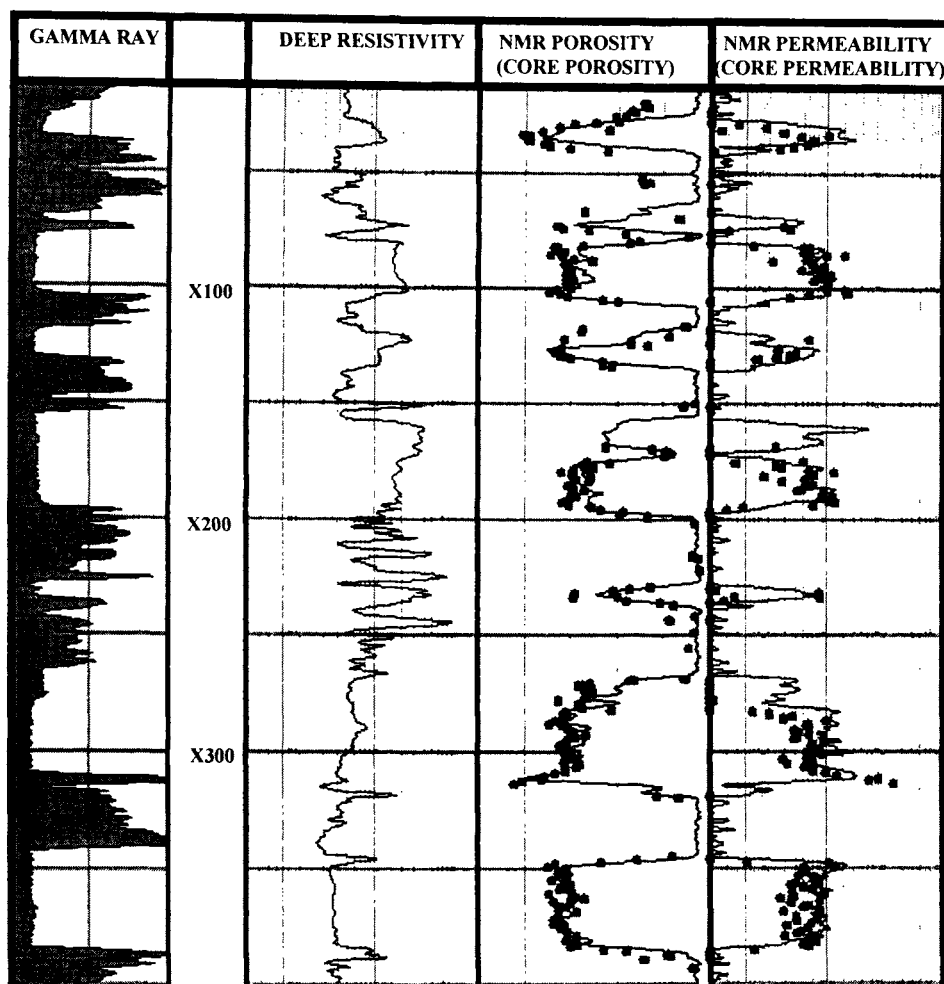


Figure 5.2: The log data showing NMR predicted permeability (curves) compares favorably with core permeabilities (solid data points). The comparison of NMR porosity estimates (curves) with core porosities (solid data points) is also shown. (From Stonard et al. [206].)

T_2 relaxation time of 750 ms, presumably the vuggy pores, is excluded, the $T_{2,LM}$ equation for permeability prediction works quite well. Chang et al. came up with a permeability formula based on laboratory NMR core measurements, i.e.,

$$k = 4.75(\phi_{NMR, < 750ms})^4 (T_{2, < 750ms})^2. \quad (5.3)$$

However, one should be aware that laboratory determined vuggy porosity T_2 cutoff may need to be modified when applied to NMR log analysis. The logging tools all have significant field gradients which will yield an apparent T_2 which in general, because of the diffusion effect, is shorter than the T_2 determined in the laboratory where the field is normally uniform.

5.4 Residual oil determination

Horkowitz et al. [103] provided a good account on using NMR logging for residual oil determination. Prior to the introduction of the current pulsed NMR logging technology, Robinson et al. [188, 189] and Setser and Williams [198] already applied the NMR inject-log technique for the residual oil saturation determination in open holes. They used dopants such as Mn-EDTA (Ethylene-Diamine-Tetra-Acetic acid) in the injected brine to kill the water signal. The presence of Mn^{++} ions shortens the relaxation time of water to below the dead time of the earth's field tool (about 30 ms²) so that only the oil signal is detected.

The advent of the new generation of pulsed NMR logging tools has greatly simplified this method. First of all, there is no need to dope the drilling mud with magnetite to kill the borehole fluid signal. Second, the reduction of the tool dead time from 30 ms to about 1 ms allows the Mn-EDTA doping procedure to measure the total porosity as well as the oil saturation. The Mn-EDTA doping provides the Mn^{++} concentration in the pore fluid needed to shorten the relaxation time of the water, to such an extent that the water signal is completely separated from the oil signal, but not too short, so that all the water signal is above the dead time and is still measurable by the tool. Figure 5.3 shows schematically such a procedure.

Paramagnetic ions dissolved in water, such as Mn^{++} and Gd^{++} , shorten the relaxation time of the hydrogen nuclei of water by the translational

²The dead time for the experimental NML was about 30 ms. The final Schlumberger tools had an effective dead time of 21 ms.

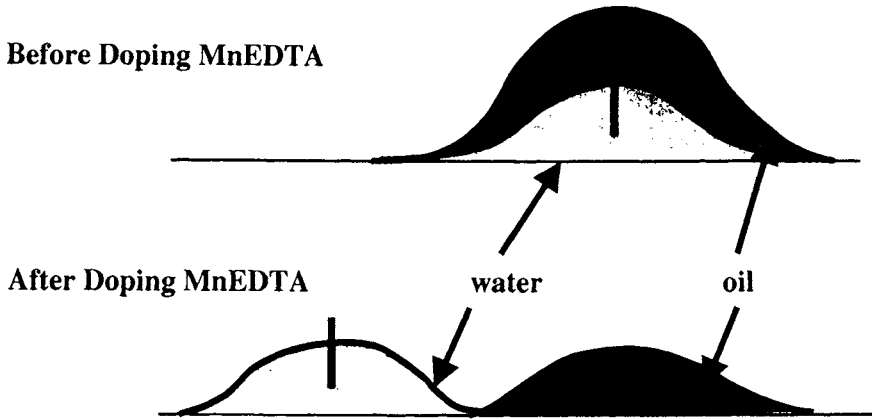


Figure 5.3: The T_2 distributions before and after Mn-EDTA doping for residual oil determination.

dipole-dipole interaction between the nuclear magnetic moment of the proton and the electronic magnetic moment of the paramagnetic ions. Bloembergen et al. [24] showed that the proton spin relaxation rate of such a solution of paramagnetic ions is given by

$$\frac{1}{T_{1,2}} = \frac{16}{15} \frac{\pi^2 \gamma^2 N_{ion} \mu^2 \eta}{kT}, \quad (5.4)$$

where γ is the proton gyromagnetic ratio, N_{ion} the ion concentration, μ the effective magnetic moment of the ion, η the viscosity, k the Boltzmann constant, and T the absolute temperature.

In addition to the dipole-dipole relaxation, the hydrogen nuclei of the water molecules also relax by spin exchange interaction with the Mn^{++} and Gd^{++} ions (Bloembergen [25]). For Mn^{++} , the spin exchange interaction is about seven times stronger than the dipole-dipole interaction.

The manganous ion is often selected for doping the drilling mud because of its high magnetic moment, good solubility, and availability. A chelating agent (such as EDTA) is frequently used to prevent Mn^{++} from forming $Mn(OH)_2$ precipitates. Thus chelation suppresses the spin exchange inter-

action between Mn^{++} and water molecules, and reduces the effect of Mn^{++} in shortening the relaxation time. Chelation also prevents ion exchange on charged surfaces such as clays in drilling mud or reservoir. As a general rule, in shaly sands or bentonite drilling muds, the Mn^{++} dopant should be chelated with EDTA to prevent flocculating the clays.

As described by Horkowitz et al. [103], the paramagnetic solution of Mn-EDTA can be prepared by mixing a 55-gal drum of Versene 100 liquid (38% Na_4EDTA) with 100 lbs of anhydrous manganous sulfate per 50 barrels of water. This results in a solution of 0.03M in manganese containing 15% excess EDTA. An excess of EDTA compared to Mn^{++} , typically 10% to 30%, is required to prevent the precipitation of 2Mn-EDTA.

Using Mn^{++} dopant in NMR logging to determine residual oil saturation apparently has been successfully applied in many different parts of the world (Horkowitz et al. [103], Crowe et al. [70], Hassoun et al. [98]). Horkowitz et al. [103] suggested using MnCl_2 instead of Mn-EDTA, with many added benefits besides reducing costs, in carbonate reservoirs where clay flocculation is not a concern.

Estimating residual oil saturation in wells drilled with oil based mud has been considered difficult or impossible. This is particularly true when using T_2 measurements to distinguish mud filtrate and low viscosity native formation oil whose T_2 contrast is small. Flaum et al. [83] suggested a fast logging procedure to provide a signal contrast when the T_1 contrast between mud filtrate and formation oil is large. Using this method, they were able to estimate residual oil saturation in the flushed zone successfully.

5.5 Hydrocarbon typing

Using NMR log to perform hydrocarbon typing was first suggested by Akkurt et al. [3], primarily for the NMR logging tools which can provide a constant field gradient, such as Halliburton/Numar's MRIL. Basically, they are using two logging parameters to obtain a contrast in the NMR signals for water, oil, and gas, i.e., the repeat delay time (variable RD) and the echo spacing (variable TE). From the analyses of the NMR data, at least qualitative hydrocarbon typing, and sometimes quantitative hydrocarbon estimates, are possible.

The method of variable TE requires a simple, well-defined field gradient. Thus, it is often limited to Halliburton/Numar's MRIL. The method of vari-

able RD, on the other hand, can be applied to both Halliburton/Numar's MRIL as well as to Schlumberger's CMR. In fact, elaborate schemes have been proposed to use the method of variable TE for hydrocarbon typing even for a complex field gradient such as that of Schlumberger's CMR, as we shall discuss later in this chapter.

The combination of variable TE and variable RD can be used not only for hydrocarbon typing but also for oil viscosity estimation [58, 59]. Recent work by Freedman et al. [90] has shown that with variable TE, variable RD, and variable field gradient, one can identify the water, oil, and gas phases, both qualitatively and quantitatively, and even provide an estimate for the oil viscosity. This is discussed in Section 5.6.

5.5.1 Multiple repeat delay times

In a standard CPMG run, a quantity called repeat delay time (or wait time, time to repeat, or RD, WT, TR for short) has to be selected. This is the length of time between any two successive CPMG echo trains. This time should be chosen to be long enough so that all spins that are tipped from the z axis into the xy plane have sufficient time to return to their original thermodynamic equilibrium state. This time is typically chosen to be 3 to 5 times the maximum T_1 relaxation time. If the RD is not long enough, then some of the tipped spins may not have fully returned to their thermal equilibrium state and a reduction of the total signal will occur, with the reduction factor given by:

$$a = 1 - e^{-RD/T_1} \quad (5.5)$$

When RD is 3 to 5 times of T_1 , the factor a , commonly known as *polarization factor*, is very close to one. If RD is smaller than T_1 , the factor a can be much less than one.

Note that both gas and oil are typically non-wetting phases. Their T_1 's can be several seconds, which are much longer than the T_1 for water in the rock (typically several hundred milliseconds). As discussed in Chapter 3, while the apparent T_2 of the oil is only slightly shifted by the field gradient, the T_2 for gas is significantly shifted to shorter relaxation times in the same gradient field due to its large diffusion coefficient. An example of T_1 and T_2 for reservoir water, oil, and gas in a field gradient of 17 Gauss/cm with a TE ($= 2\tau$) of 1.2 ms is shown in Table 5.1.

Table 5.1: An example of T_1 and T_2 of reservoir fluids in a field gradient of 17 Gauss/cm for a typical sandstone reservoir, where the TE ($= \tau$) is 1.2 ms.

	T_1 ms	D $10^{-5}\text{cm}^2/\text{s}$	$\frac{1}{3}\gamma^2 G^2 D \tau^2$ s^{-1}	$T_{2,\text{apparent}}$ ms
Brine	1-500	7.7	1.9	0.67-200
Oil	1000	2	0.5	670
Gas	4000	100	24.8	40

Thus the T_2 distribution for the reservoir rock containing water, oil, and gas could have a general form as illustrated schematically in Figure 5.4. One can have two logging passes with a RD_S shorter than $T_{1,\text{gas}}$ for gas, or $T_{1,\text{oil}}$ for oil, but long enough for the longest T_{1w} component of the water in the rock to recover, and another RD_L long enough for capturing all gas and oil signals as follows:

$$3T_{1w,\text{max}} < RD_S < T_{1,\text{gas}} \text{ (and/or } T_{1,\text{oil}}) < RD_L. \quad (5.6)$$

Then by subtracting the pass with the short wait time, RD_S , from that with the long wait time, RD_L , the portions for the water signal cancel out, leaving only the contributions from the gas and oil, as shown in the following:

$$\phi_{RD_L} = \phi_w + \phi_{gas}(1 - e^{-RD_L/T_{1,gas}}) + \phi_{oil}(1 - e^{-RD_L/T_{1,oil}}) \quad (5.7)$$

$$\phi_{RD_S} = \phi_w + \phi_{gas}(1 - e^{-RD_S/T_{1,gas}}) + \phi_{oil}(1 - e^{-RD_S/T_{1,oil}}) \quad (5.8)$$

$$\phi_{RD_L} - \phi_{RD_S} = \phi_{gas}e^{-RD_S/T_{1,gas}} + \phi_{oil}e^{-RD_S/T_{1,oil}} \quad (5.9)$$

where ϕ_{RD_L} and ϕ_{RD_S} represent the T_2 signals for long and short repeat delay times, respectively, and ϕ_w the water signal, ϕ_{gas} the gas signal, and ϕ_{oil} the oil signal. The factors, $e^{-RD_L/T_{1,gas}}$ and $e^{-RD_L/T_{1,oil}}$, presumably are very small and can be dropped out.

From Eq. (5.9), one should be able to obtain the gas and oil saturations, assuming the hydrogen index for the gas is known. This is the basis for

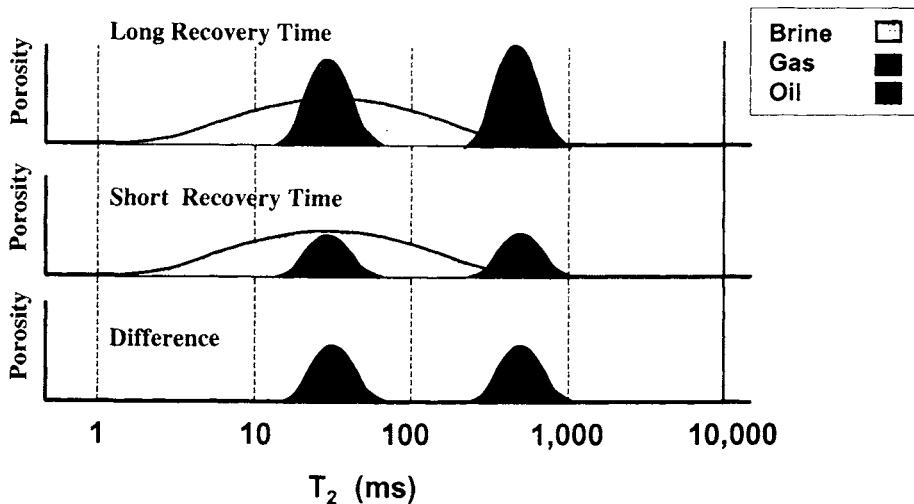


Figure 5.4: Schematic diagram of T_2 distributions for a reservoir rock containing water, oil, and gas in a differential spectrum procedure.

the multiple repeat delay time method, also known as Differential Spectrum Method (DSM).

In practice, subtracting the T_2 distribution of the short repeat delay time from that of the long repeat delay time usually does not work very well. This is because the typical noise level in the NMR logging measurements is quite high, and there are uncertainties of the same order of magnitude as the answer we are seeking in the T_2 distributions.

Prammer et al. [178] extended Akkurt et al.'s method [3] by doing this in the time domain, i.e., subtracting the echo train of the short repeat delay time from that of the long repeat delay time, and inverting the difference of the echo trains to obtain the T_2 distribution. This is called the *time domain analysis* (TDA). This approach is more reliable than the subtraction in the T_2 domain, because it bypasses the issue of uncertainty associated with the inversion. To obtain reliable results, frequently a significant stacking of the data of many depth intervals is required. In order to have a meaningful outcome, a relatively large volume of high pressure gas is required to have a discernible difference between the two logging passes with different RD's.

5.5.2 Variable echo spacing

There is another parameter in the CPMG T_2 measurements which we can manipulate to extract more information, i.e., the echo spacing, or the time between echoes TE. As described in Chapter 2, the T_2 relaxation rate of bulk water in a constant field gradient is modified as follows:

$$\text{Rate} = \frac{1}{T_{2B,w}} + \frac{1}{3}\gamma^2 G^2 D_w \tau^2, \quad (5.10)$$

where $T_{2B,w}$ is the T_2 for bulk water, G the constant field gradient, and D_w the self diffusion coefficient of water. For porous media such as rocks in a constant field gradient, Eq. (5.10) is further modified as:

$$\text{Rate} = \frac{1}{T_{2B,w}} + \frac{1}{T_2} + \frac{1}{3}\gamma^2 G^2 D_w \tau^2, \quad (5.11)$$

where second term is related to the surface to volume ratio:

$$\frac{1}{T_2} = \rho \frac{S}{V}, \quad (5.12)$$

and the first term is negligible compared to the second term. Thus, when we have a fluid-saturated rock with a wide distribution of pore sizes in a constant field gradient which is larger than internal gradients from susceptibility differences, at times short enough that most fluid molecules have not hit pore walls or phase boundaries, we would expect each relaxation rate $1/T_{2j}$ to increase by an amount of

$$\frac{1}{3}\gamma^2 G^2 D_w \tau^2, \quad (5.13)$$

where $\tau = \text{TE}/2$. Hence, by increasing TE, we would expect the T_2 amplitude distributions to shift to shorter relaxation times. Since the increase in the relaxation rate is a constant, the shift of T_2 due to this diffusion term is larger at long relaxation times but quite small at short relaxation times. For example, if the diffusion term $\gamma^2 G^2 D_w \tau^2$ is 2s^{-1} , then the relaxation time at 500 ms would shift to 250 ms, whereas the relaxation time at 5 ms would only shift to 4.95 ms.

Because oil is often the non-wetting phase, its relaxation rate, or $1/T_{2,o}^\dagger$ (where $T_{2,o}^\dagger$ is the apparent $T_{2,o}$ value in the constant field gradient, and we

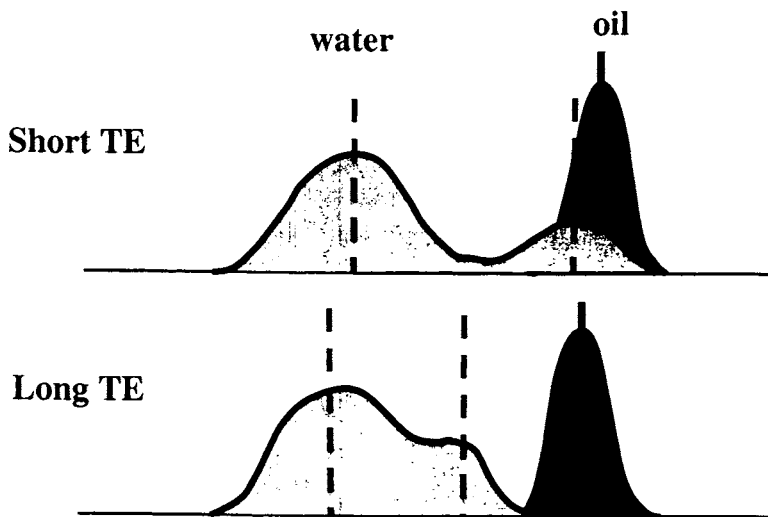


Figure 5.5: In the shift spectrum scheme, the T_2 distribution of a long TE helps resolve water and oil signals that were overlapping in the T_2 distribution of a short TE run.

use the subscript “o” to represent oil), depends on the bulk $T_{2B,o}$, its diffusion coefficients D_o , the gradient strength G , and the half echo spacing τ , i.e.;

$$\frac{1}{T_{2,o}^\dagger} = \frac{1}{T_{2B,o}} + \frac{1}{3}\gamma^2 G^2 D_o \tau^2 \quad (5.14)$$

The diffusion coefficient of oil, D_o , is usually an order of magnitude less than that of water. Thus, the diffusion term in the relaxation rate equation (Eqs. (5.11) and (5.14)) for water is much larger than that for oil. As a result, as the TE is increased, the shift of T_2 for water is much more than that for oil. Therefore, if the signals for water and oil overlap initially, a larger TE can often produce enough relative shift in T_2 and hence resolve the water and the oil signal. This is usually quite effective for long T_2 relaxation times (several hundred milliseconds), because the shift there is quite significant. Figure 5.5 shows schematically such a shift in T_2 peaks of water which helps resolve the oil signal.

The diffusion coefficient of gas is about two orders of magnitude larger than that of water. The apparent T_2 for gas for a field gradient of 17

Gauss/cm and a TE of 1.2 ms is already in the neighborhood of 40-60 ms. Further increase in TE would shift the gas signal into the region of irreducible water. Besides, the gas signal is very weak, and it is very hard to detect the effect of varying TE. Hence, the *Shift Spectrum Method* is commonly used for detecting oil, whereas the *Differential Spectrum Method* can be used for both gas and oil.

The variable TE method or the *Shift Spectrum Method* has been used by many to extract oil saturation from NMR logs. Looyestijn [145] used the large contrast of diffusion coefficients between water and oil and NMR logs taken with different echo spacings to solve for the oil saturation with reasonable success. Slijkerman et al. [203] used forward modeling of a multifluid relaxation model to solve for oil and water saturations.

5.5.3 Enhanced diffusion

The method of variable wait time relies on a large T_1 contrast between hydrocarbon and water. When such contrast becomes small, e.g., when the T_1 of the oil is close to that of the water, this method is not effective. If however, the diffusion coefficient of the oil is still reasonably larger than that of the water, then with a large enough TE, enough contrast in the T_2 for oil and water can still be produced. Based on this, Akkurt et al. suggested [4, 5] a single large TE (4.8 or 6 ms) for a dual wait time run to resolve this problem. They called this *Enhanced Diffusion Method* (EDM). Figure 5.6 shows schematically the general concept of the procedure.

Recall that the apparent T_2 for water in a gradient field is given by

$$\begin{aligned} \frac{1}{T_2^{\dagger}} &= \frac{1}{T_{2B,w}} + \frac{1}{T_{2,surf}} + \frac{1}{3}\gamma^2 G^2 D_w \tau^2 \\ &= \frac{1}{T_{2,int}} + \frac{1}{T_{2D}} \end{aligned} \quad (5.15)$$

where the first term on the right hand side, $1/T_{2B,w}$ (the bulk relaxation rate), can often be neglected; the second term, $1/T_{2,surf}$ (the surface relaxation rate), when combined with the first term, can be designated as the intrinsic relaxation rate, $1/T_{2,int}$; and the last term, $1/T_{2D}$, is defined as

$$T_{2D} = 3(\gamma^2 G^2 D_w \tau^2)^{-1} \quad (5.16)$$

A key point, brought forward by Akkurt et al. [4, 5], is that T_{2D} is the upper bound of all the water signal. This can be easily appreciated by consid-

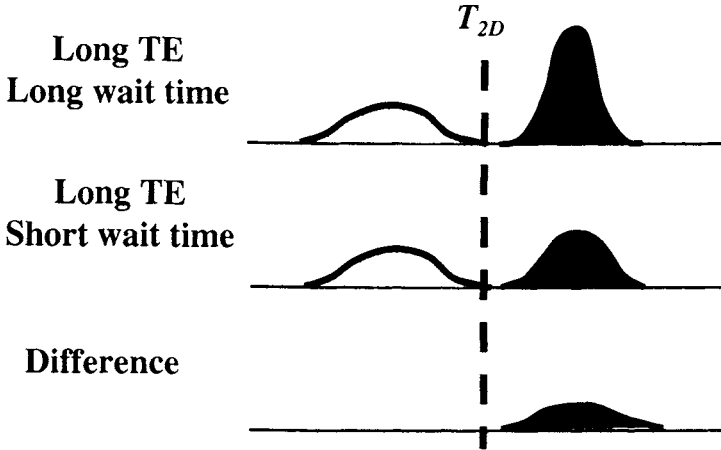


Figure 5.6: The T_2 distribution of long TE for dual wait times can help remove the water signal in an enhanced diffusion run. The T_{2D} is the upper bound for the water signal.

ering Eq. (5.15). For all the possible values that $T_{2,int}$ can have, the apparent $T_2^\dagger$ will always be less than T_{2D} . If $T_{2,int}$ is small, then $T_2^\dagger \approx T_{2,int}$. When $T_{2,int}$ tends to infinity, then $T_2^\dagger \approx T_{2D}$, with T_{2D} being its upper limit. Thus, when a large TE is used, T_{2D} shifts toward shorter relaxation times. Any signal beyond T_{2D} must then come from oil.

This becomes a simple way to judge the nature of the signal beyond T_{2D} , i.e., it must come from oil. However, it does not preclude the possibility that part of the signal below T_{2D} can also come from oil, or that all oil signal happens to be less than T_{2D} . Thus, Enhanced Diffusion Method (EDM) allows you to make a statement about some of the observations, but does not provide you with a complete picture unless you have more information, such as laboratory NMR measurements of the produced oil of the reservoir. Other issues such as the noise level of the signal can also cloud the judgment about T_2 components beyond T_{2D} . The regularization process³ of noisy data can often cause water signal to spill over the T_{2D} limit.

³Regularization is a technique of NMR data inversion which is discussed in Chapter 6.

With the laboratory information of the produced oil, as well as the dual wait time data set, it is possible to perform elaborate time domain analysis to extract a more accurate estimate of the oil saturation.

5.5.4 Complex field gradients

Our discussion so far on the variable RD and TE methods are mostly used for NMR tools which have a simple, single-valued, and well-defined constant field gradient, such as Halliburton/Numar's MRIL. For Schlumberger's CMR, which has a complicated field gradient distribution, implementation of these methods is not as straightforward. Generally speaking, the variable RD method requires a good T_1 contrast and is not related to the field gradient. Hence, it can be applied to both MRIL and CMR. On the other hand, the variable TE method requires a well-defined field gradient, and is often limited to tools such as MRIL. However, it is possible to implement the variable TE method for tools with complex field gradients. Elaborate schemes have been developed for CMR and tested with reasonable success.

Flaum et al. [82] suggested methods to determine the gas saturation, using NMR signal amplitude, echo difference, and echo ratio (variable TE). Freedman et al. [89] later proposed a formula using the density porosity and the NMR porosity to compute the gas saturation. Flaum et al.'s method [84] was recently extended, using variable TE, to estimate the oil saturation with reasonably good results [84].

Gas saturation

Flaum et al.'s NMR signal amplitude method [82] for the gas saturation determination is quite similar to Freedman et al.'s approach [89]. The latter provides a more extensive formulation, which we shall discuss as follows.

This method requires a good knowledge of the grain density and a good quality density porosity. Combining these with a knowledge of the hydrogen index and the T_1 value for gas, the gas saturation can be accurately determined.

Recall that the bulk density ρ_b is given by

$$\rho_b = (1 - \phi_{den})\rho_m + \rho_f\phi_{den}, \quad (5.17)$$

where ρ_m and ρ_f are the matrix and fluid densities, respectively, and the

density porosity ϕ_{den} is

$$\phi_{den} = \frac{\rho_m - \rho_b}{\rho_m - \rho_f}. \quad (5.18)$$

When there is gas in the formation, Eq. (5.17) becomes

$$\rho_b = (1 - \phi)\rho_m + \rho_f(1 - S_g)\phi + \rho_g S_g \phi, \quad (5.19)$$

where S_g is the gas saturation in fractions, ρ_g is the gas density, and ϕ is the true total porosity. The ϕ_{NMR} can be written as follows:

$$\phi_{NMR} = \phi(1 - S_g) + S_g \phi \cdot HI \cdot P, \quad (5.20)$$

where HI is the hydrogen index, and P is the polarization factor

$$P = 1 - e^{-RD/T_1}, \quad (5.21)$$

where RD is the repeat delay time. Combining Eqs. (5.17), (5.19), and (5.20) to eliminate ϕ , we finally get

$$S_g = \frac{\phi_{den} - \phi_{NMR}}{\phi_{den}(1 - HI \cdot P) + \lambda \cdot \phi_{NMR}} \quad (5.22)$$

for the gas saturation, S_g , and

$$\phi = \frac{\phi_{den}(1 - HI \cdot P) + \lambda \cdot \phi_{NMR}}{(1 - HI \cdot P) + \lambda} \quad (5.23)$$

for the true total porosity, ϕ ; where $\lambda = (\rho_f - \rho_g)/(\rho_m - \rho_f)$.

Freedman et al.'s method is not limited to tools with complex field gradients, and hence is applicable to MRIL as well. The approach of echo ratio proposed by Flaum et al. [82] for gas zone identification is specifically for tools with complex field gradients such as the CMR. It involves mapping out the field gradient distribution for application to the echo ratio analysis. Since it was further extended to solve for oil saturation, we shall combine the discussion in the following section.

Oil saturation

Flaum et al.'s method for gas/oil saturation determination [82, 84] utilizes the variable TE approach in a complex field gradient. To proceed, we should

first understand how the T_2 echo train is affected in a complex field gradient. We know that for a constant field gradient, G , the amplitudes of a spin echo train can be written as:

$$A_i(t_i, \tau, D) = \sum_{j=1}^m f_j e^{-t_i/T_j} e^{-t_i/T_{2D}}, \quad (5.24)$$

where A_i is the amplitude of the i -th echo, f_j is the partial porosity whose spins have a T_2 relaxation time of T_j ($j = 1, \dots, m$, and there are m relaxation times), and T_{2D} is the constant gradient term as defined previously (Eq. (5.16)):

$$T_{2D} = 3(\gamma^2 G^2 D \tau^2)^{-1}. \quad (5.25)$$

If we make the assumption that the sample can be subdivided into small volume elements, each of which has a more or less constant field gradient, and that all these contributions can be integrated, then we have the echo amplitude as

$$A_i(t_i, \tau, D) = \int J(G) \sum_{j=1}^m f_j e^{-t_i/T_j} e^{-t_i/T_{2D}} dG \quad (5.26)$$

where $J(G)dG$ is the infinitesimal portion which has a gradient of G . Note this assumption is only an approximation. There are some intricate details about the diffusion of spins in a non-uniform gradient field, which we shall discuss in the last chapter.

Neglecting any susceptibility contrast induced internal field gradient, we can place the integration inside the summation:

$$A_i(t_i, \tau, D) = \sum_{j=1}^m f_j e^{-t_i/T_j} \int J(G) e^{-t_i/T_{2D}} dG \quad (5.27)$$

To obtain the field gradient distribution $J(G)$, we can use a bottle of water sample whose $T_{2B,w}$ is known accurately beforehand. Since this is bulk water, the summation of various T_j reduces to a single term, $e^{-t_i/T_{2B,w}}$. Dividing both sides of Eq. (5.27) by $e^{-t_i/T_{2B,w}}$, we get

$$\tilde{A}_i(t_i, \tau, D) = \frac{A_i}{e^{-t_i/T_{2B,w}}} = \int J(G) e^{-t_i/T_{2D,w}} dG. \quad (5.28)$$

Upon close examination of Eq. (5.28), it is easily realized that we now have a problem very similar to inverting an echo train to get a T_2 amplitude

distribution. The “echo” amplitude is $\tilde{A}_i$, the “ T_2 ” is $T_{2D,w}$ (which is a function of G as shown in Eq. (5.16)), and the “ T_2 amplitude” is simply $J(G)$.

Once $J(G)$ is found, it is straightforward to establish the following function $F(t_i, \tau, D)$ for any fluid or gas with a different diffusion constant, D , and hence, T_{2D} ,

$$F(t_i, \tau, D) = \int J(G) e^{-t_i/T_{2D}} dG. \quad (5.29)$$

To determine the gas zone, Flaum et al. [82] tried to extract information of the apparent diffusion constant, D , of the formation fluid from the NMR logs. As the diffusion constant becomes significantly higher than that of the water, it is taken as a sign of entering the gas zone.

The echo ratio method was used to determine the apparent D . The ratio of echo amplitudes taken at the same elapsed time t for two different echo trains measured with a short and a long τ is given as follows:

$$\frac{A_i(t_i, \tau_{short}, D)}{A_i(t_i, \tau_{long}, D)} = \frac{F(t_i, \tau_{short}, D)}{F(t_i, \tau_{long}, D)}, \quad (5.30)$$

where the summation part before the integral in the RHS of Eq. (5.27) has been canceled.

Flaum et al. [82] found empirically that the apparent D is given by

$$D = aM^2 + bM, \quad (5.31)$$

where M is the first moment of the echo amplitude ratio distribution minus 1:

$$M = \sum_i t_i [F(t_i, \tau_{short}, D)/F(t_i, \tau_{long}, D) - 1]. \quad (5.32)$$

The constants a and b can be determined beforehand with the bottle of water experiment. The ratio of echo amplitudes was calculated using the fitted decay curves to avoid the noise problem. Flaum et al. [82] found that the method provided a good prediction of the gas zone, although the extracted apparent D for the gas was about a factor 3 less than it should be in downhole conditions. Considering the restricted diffusion effect and the presence of water and oil, such a reduction is not unexpected.

To determine the oil saturation, Flaum et al. [84] suggested a combination of time domain analysis and the variable TE method, which they called *Enhanced Diffusion Analysis* (EDA).

The difference of two echo trains with different TE is obtained as follows:

$$\begin{aligned}
 \Delta A_i &= A_i(t_i, \tau_{short}, D) - A_i(t_i, \tau_{long}, D) \\
 &= \sum_{j=1}^m f_j e^{-t_i/T_j} [F(t_i, \tau_{short}, D_w) - F(t_i, \tau_{long}, D_w)] \\
 &\quad + f_{oil}(1 - e^{-RD/T_{1,oil}}) e^{-t_i/T_{2,oil}} [F(t_i, \tau_{short}, D_{oil}) - F(t_i, \tau_{long}, D_{oil})]
 \end{aligned} \tag{5.33}$$

where RD is the repeat delay time used in the acquisition of the echo trains.

The actual procedure for carrying out the *Enhanced Diffusion Analysis* is somewhat involved. Eq. (5.33) is first solved in the water zone for the T_2 signal of water by assuming a known D_w . This is accomplished through trial and error by matching the echo difference to the predicted water signal. Then the same shape of the T_2 distribution for water is assumed in the oil zone to solve for the oil saturation using Eq. (5.33). As some of the water residing in the small pores does not contribute to the water signal in the difference of the two echo trains due to restricted diffusion, proper steps should be taken to take this into consideration. Flaum et al. [84] have shown that this method can be successfully carried out in a finite interval where the rock fabrics are similar between the water and oil zones.

Thus, we see that through an elaborate and complicated scheme, the variable TE method can also be implemented for a tool with complex field gradients. The physics involved, however, is not as straightforward as that for a constant field gradient.

5.6 Oil viscosity estimation

The NMR log is the only log so far that provides information on the oil viscosity. As we discussed in Chapter 4, the T_1 or T_2 of oil is related to viscosity, η , through the following relationship (Vinegar [220]):

$$T_{1,2} = \frac{C_1 T_K}{298\eta}, \tag{5.34}$$

where $T_{1,2}$ is in seconds, and usually refers to the logarithmic average of $T_{1,2}$; T_K , the absolute temperature, is in degrees Kelvin; η , the viscosity, is in centipoise (cp); and C_1 is an empirical fitting constant whose value varies

from 2.8 to 1.2 depending upon whether the oil sample is oxygen free or not (see discussion in Chapter 3).

The diffusion constant of the oil is also related to the viscosity (Vinegar [220]):

$$D = \frac{C_2 T_K}{298 \eta} \quad (5.35)$$

where D is in units of $10^{-5} \text{cm}^2/\text{sec}$, T_K and η are in the same units as described previously, and C_2 is approximately 1.4 for oil and 2 for water, respectively.

As a first order approximation, if one can identify the T_2 peak related to the oil signal, one can simply use Eq. (5.34) directly to estimate the oil viscosity. However, after some thought, one should realize that Eq. (5.34) is obtained in the laboratory with a homogeneous magnetic field, whereas the apparent T_2 peak obtained from the NMR logs is most likely affected by the magnetic field gradient G as described in the following:

$$\begin{aligned} \frac{1}{T_2^\dagger} &= \frac{1}{T_{2,int}} + \frac{1}{T_{2D}} \\ &= \frac{1}{T_{2,int}} + \frac{1}{3}(\gamma^2 G^2 D \tau^2) \end{aligned} \quad (5.36)$$

Thus for large viscosity, or small diffusion constant, and when the contribution from the diffusion term $1/T_{2D}$ is small, the first order approximation of ignoring the diffusion contribution and using Eq. (5.34) directly would be acceptable. If this is not satisfied, then more elaborate analysis is necessary.

5.6.1 Multiple TEs and repeat delay times

Using a combination of multiple TE and dual wait time scheme, Chen et al. [58, 59] suggested a method which provides quite successful estimates of oil viscosity from the NMR logs. Their procedure consists of the following three steps:

1. Separate the oil signal from the water signal using time domain analysis of a dual wait time run.
2. Determine the apparent T_2 for oil for different TE runs, assuming it is the non-wetting phase.

3. Estimate the oil viscosity using $T_{2,int}$ or T_{2D} .

The first step is accomplished by conducting three different TE (2τ) passes, each with a dual wait time, i.e., a long and a short wait time. The echo train of the short wait time is subtracted from that of the long wait time in the time domain. The difference of the echo trains is assumed to be primarily coming from the oil signal for the oil bearing zone. It can be inverted to get a T_2 distribution, primarily for oil with a unimodal shape, or it can be fitted with a single exponential. To ensure the residual water signal in the difference of the echo trains is removed, the data of the first 24 ms is omitted in the processing. The apparent T_2 is then determined for each different TE. This completes the second step. The last step is to plot the reciprocal of this apparent T_2 , or apparent relaxation rate, as a function of τ^2 :

$$\begin{aligned}\frac{1}{T_2^\dagger} &= \frac{1}{T_{2,int}} + \frac{1}{3}(\gamma^2 G^2 D)\tau^2 \\ &= A + B\tau^2\end{aligned}\quad (5.37)$$

where the intercept A , determined from the linear regression fit, gives $1/T_{2,int}$. For the non-wetting phase, it is the same as the bulk value, and hence, Eq. (5.34) can be used to estimate the viscosity. The slope B , of such a fit, gives a diffusion constant D , provided the gradient G is known precisely. From the value of D and Eq. (5.35), the oil viscosity can also be evaluated. Thus we have two independent methods to estimate oil viscosity, i.e., one from the intercept and Eq. (5.34), and the other from the slope and Eq. (5.35). These two estimates can be used as a cross check of the procedure.

When there are strong internal field gradients due to the presence of ferro- and paramagnetic materials, the value of G may be significantly modified. In this case, the oil viscosity derived from the intercept is more reliable.

The oil viscosity η can also be obtained directly from a nonlinear fit of the following equation (derived by combining Eqs. (5.34), (5.35), and (5.37)):

$$\frac{1}{T_2^\dagger} = A_1 \cdot \eta + A_2 \tau^2 \cdot \eta^{-1} \quad (5.38)$$

where $A_1 = 298/(C_1 T_K)$ and $A_2 = \gamma^2 G^2 C_2 T_K / (298 \cdot 3)$. Naturally, in this approach, you have to assume that G is accurately known and the internal gradient is insignificant.

5.6.2 Summation of echoes

The summation of echoes was first suggested by Sezginer et al. [199] as a permeability indicator. Later work by Chen et al. [60] showed that the summation of echoes is a very useful petrophysical parameter which can be used for estimating the T_1 of oil, hence for oil viscosity determination, as well as depth-matching, log quality control, etc.

To understand what kind of information is contained in the summation of echoes, let us first examine this quantity analytically. The echo amplitude, g_i , for a T_2 CPMG measurement is typically expressed as:

$$g_i = \sum_{j=1}^m f_j e^{-it_E/T_j} + \epsilon_i \quad (5.39)$$

where f_j is the partial porosity for T_2 relaxation time $T_{2,j}$ (There are m relaxation times, $j = 1, \dots, m$, and $\sum_j f_j = \phi$, where ϕ is the total porosity), t_E is TE, and ϵ_i is the noise for the i -th echo amplitude. Thus, if we denote the summation of echoes by SE , we get,

$$\begin{aligned} SE &= \sum_{i=1}^n \sum_{j=1}^m f_j e^{-it_E/T_{2,j}} + \sum_{i=1}^n \epsilon_i \\ &= \sum_{j=1}^m f_j \frac{1 - e^{-nt_E/T_{2,j}}}{e^{t_E/T_{2,j}} - 1} + \sum_{i=1}^n \epsilon_i \\ &\approx \sum_{j=1}^m f_j \frac{T_{2,j}}{t_E} + \mathcal{O}[\sqrt{n}\sigma] \\ &= \frac{\phi \langle T_2 \rangle}{t_E} + \mathcal{O}[\sqrt{n}\sigma] \end{aligned} \quad (5.40)$$

where σ is the statistical average of the noise, and we have made the following assumptions

$$t_E \ll T_{2,j} \longrightarrow e^{t_E/T_{2,j}} \approx 1 + \frac{t_E}{T_{2,j}} \quad (5.41)$$

$$\text{and} \quad nt_E \gg T_{2,j} \longrightarrow e^{-nt_E/T_{2,j}} \approx 0. \quad (5.42)$$

Thus we can immediately see the benefit of this method that the summation of echoes is proportional to the porosity times the average of the T_2 . The simple average of T_2 is weighted heavily toward the long relaxation

times. This is the reason why the summation of echoes can be used as a permeability indicator. However, there are situations when $\langle T_2 \rangle$ does not correlate well with permeability.

To apply the summation of echoes to estimate the T_1 of the oil, two NMR logs with different wait times should be performed. We note that in the water and oil zone, the echo amplitudes for long and short wait times, $g_{i,L}$ and $g_{i,S}$, are given respectively as:

$$\begin{aligned} g_{i,L} &= \sum_{j=1}^m f_{j,w} e^{-it_E/T_{2,j}} + \phi_o H I_o \alpha_L e^{-it_E/T_{2,o}} \\ g_{i,S} &= \sum_{j=1}^m f_{j,w} e^{-it_E/T_{2,j}} + \phi_o H I_o \alpha_S e^{-it_E/T_{2,o}} \end{aligned} \quad (5.43)$$

where ϕ_o is the porosity associated with oil, $H I_o$ is the hydrogen index for oil, and $\alpha_L = 1 - e^{-RD_L/T_{1,o}}$ and $\alpha_S = 1 - e^{-RD_S/T_{1,o}}$ are polarization factors for long and short wait times for oil, respectively. Thus the corresponding summations of echoes for long and short wait times are:

$$\begin{aligned} SE_L &\approx \frac{\phi_w \langle T_{2,w} \rangle}{t_E} + \frac{H I_o \alpha_L \phi_o T_{2,o}}{t_E} (1 - e^{-nt_E/T_{2,o}}) \\ SE_S &\approx \frac{\phi_w \langle T_{2,w} \rangle}{t_E} + \frac{H I_o \alpha_S \phi_o T_{2,o}}{t_E} (1 - e^{-nt_E/T_{2,o}}) \end{aligned} \quad (5.44)$$

If we assume that the first term for the water, i.e.,

$$f_w = \frac{\phi_w \langle T_{2,w} \rangle}{t_E} \quad (5.45)$$

is known from the T_2 distributions obtained from the regular inversion, then the ratio

$$R = \frac{SE_L - f_w}{SE_S - f_w} = \frac{\alpha_L}{\alpha_S} = \frac{1 - e^{-RD_L/T_{1,o}}}{1 - e^{-RD_S/T_{1,o}}} \quad (5.46)$$

can be used to estimate T_1 of the oil.

This method is somewhat better than the regular time domain analysis. The latter requires prior knowledge of the tool field gradient to relate $T_{1,o}$ and $T_{2,o}$, and needs to solve for ϕ_o and $T_{1,o}$ simultaneously. The process is also strongly affected by the noise level and the different methods of regularization.

The summation of echoes method, on the other hand, needs to solve for only one parameter, T_1 , thus providing a good estimate of oil viscosity. The quantities, SE_L and SE_S have been determined with very high precision, resulting from the effective reduction of noise in the time domain. Their difference is enhanced by the factor $T_{2,o}(1 - e^{-nt_E/T_{2,o}})/t_E$. Furthermore, the inaccuracy in the quantity for water, f_w , is relatively unimportant, because $\langle T_{2,w} \rangle \ll T_{2,o}$.

Because of the high precision of SE , it can be used for depth matching, especially for passes with high noise level where depth matching is difficult. Note in Eq. (5.44), because $\langle T_{2,w} \rangle \ll T_{2,o}$, the value of SE is quite insensitive to the first term, but very sensitive to the second term. Thus it can also be used for log quality control to determine whether the wait time is sufficient in the logging passes.

5.6.3 Constituent viscosity model

Freedman et al. [90] recently suggested a pore fluids characterization method using a suite of spin-echo measurements with different echo spacings, polarization times, applied field gradients, and number of echoes to invert the T_2 distributions of various pore fluids. They called this, *magnetic resonance fluid* (MRF) characterization method.

Freedman et al.'s approach is much more ambitious than Chen et al.'s method. Not only does it yield information on the oil viscosity, it also yields the T_2 distribution of oil, hence the oil saturation, as well as the gas saturation, if there is any. It uses six or more measurements for one sample or one formation interval to solve for all unknowns in the amplitudes of the T_2 distributions for water, oil, and gas (plus oil-based mud filtrate, if present). Thus, unlike Chen et al.'s approach where only a single T_2 is used for oil, Freedman et al.'s method provides a T_2 distribution for oil. This makes a significant difference when the T_2 distribution for oil is multi-modal rather than unimodal which can be represented by a single T_2 .

Freedman et al.'s method has not been widely tested yet. The ability of their method to invert the relative amplitudes for overlapping water and oil signals hinges upon the important hypothesis of the *Constituent Viscosity Model* (CVM). This CVM hypothesis links the constituents of the $T_{1,2}$ relaxation times of a crude oil with those of the diffusion coefficients of the same crude oil, as described in the following.

If the macroscopic empirical correlation of T_2 (or more specifically $T_{2,LM}$,

the logarithmic average of T_2) is given by the following equation:

$$T_{2,LM} = \frac{aT_K}{\eta} \quad (5.47)$$

where $T_{2,LM}$ is the logarithmic average over $T_{2,j}$:

$$\log T_{2,LM} \equiv \sum_{j=1}^m f_j \log T_{2,j}, \quad \sum_{j=1}^m f_j = 1, \quad (5.48)$$

then the *Constituent Viscosity Model* defines a constituent viscosity, η_j , given by the following relationship,

$$T_{2,j} = \frac{aT_K}{\eta_j} \quad (5.49)$$

with the natural consequence:

$$\log \eta_{LM} \equiv \sum_{j=1}^m f_j \log \eta_j = \log \eta. \quad (5.50)$$

Note that the constituent viscosity, η_j , is strictly and operationally defined by Eq. (5.49), and has no real physical meaning except to indicate some kind of smeared out distribution of the components of a mixture. For example, a crude oil could be a mixture of three components, while its T_2 distribution could have many $T_{2,j}$ relaxation times, and thus as many η_j . There could be as many as, say, 20 η_j , whereas there are only three true physical components in the mixture.

The critical hypothesis that CVM makes is the following: In a Stejskal and Tanner type pulsed field gradient spin-echo experiment (PFGSE) [205, 209], the constituent diffusion coefficients, D_j , would have exactly the same amplitude, f_j , from the inversion of the following signal ratios obtained from the PFGSE experiments:

$$\frac{M}{M_0} = \sum_{j=1}^m f_j e^{-\gamma \delta^2 G^2 (\Delta - \frac{1}{3}\delta) D_j} \quad (5.51)$$

as those in Eq. (5.48), if the D_j is chosen in the following manner

$$D_j = \frac{bT_K}{\eta_j}. \quad (5.52)$$

With this hypothesis, it easily follows that

$$\eta = \frac{aT_K}{T_{2,LM}} = \frac{bT_K}{D_{LM}} \equiv \eta_{LM} \quad (5.53)$$

where

$$\log D_{LM} \equiv \sum_{j=1}^m f_j \log D_j, \quad (5.54)$$

and that

$$\frac{D_j}{T_{2,j}} = \frac{b}{a} = \frac{D_{LM}}{T_{2,LM}}. \quad (5.55)$$

Thus an important consequence of this CVM hypothesis is that the T_2 and D distributions are not independent; i.e., if one is known, the other one can be calculated. The CVM was mainly based on the experimental work of Lo et al. [144]. There is some experimental evidence to support the CVM hypothesis which links the T_2 and D distributions, but it has not been extensively tested.

With the CVM hypothesis, Freedman et al. were able to successfully invert the T_2 amplitudes of water, oil, gas, and oil-based mud filtrate (OBMF) using the following multifluid relaxational model:

$$\begin{aligned} g_i^p &= \sum_{j=1}^m f_{w,j} e^{-it_{E_p}/T_{2w,j}^\dagger(p)} (1 - e^{-RD_p/\xi T_{2,j}}) \\ &+ \sum_{j=1}^m f_{o,j} e^{-it_{E_p}/T_{2o,j}^\dagger(p)} (1 - e^{-RD_p/T_{1,o}}) \\ &+ f_g e^{-it_{E_p}/T_{2g}^\dagger(p)} (1 - e^{-RD_p/T_{1,g}}) \\ &+ f_{OBMF} e^{-it_{E_p}/T_{2,OBMF}^\dagger(p)} (1 - e^{-RD_p/T_{1,OBMF}}) \end{aligned} \quad (5.56)$$

where g_i^p is the i -th echo of the p -th experiment, t_{E_p} the echo spacing, RD_p the repeat delay time, ξ the T_1/T_2 ratio for water, and $f_{w,j}$, $f_{o,j}$, f_g and f_{OBMF} the partial amplitudes for water, oil, gas, and oil-based mud filtrate (OBMF), respectively. The apparent T_2 's for water ($T_{2w,j}^\dagger(p)$), oil ($T_{2o,j}^\dagger(p)$), gas ($T_{2g}^\dagger(p)$), and OBMF ($T_{2,OBMF}^\dagger(p)$) are given as follows:

$$\frac{1}{T_{2w,j}^\dagger(p)} = \frac{1}{T_{2w,j}} + \frac{1}{12} \gamma^2 G_p^2 t_{E_p}^2 D_w \quad (5.57)$$

$$\frac{1}{T_{2o,j}^\dagger(p)} = \frac{1}{T_{2o,j}} + \frac{1}{12} \gamma^2 G_p^2 t_{E_p}^2 D_o \quad (5.58)$$

$$\frac{1}{T_{2g}^\dagger(p)} = \frac{1}{T_{1,g}} + \frac{1}{12} \gamma^2 G_p^2 t_{E_p}^2 D_g \quad (5.59)$$

$$\frac{1}{T_{2,OBMF}^\dagger(p)} = \frac{1}{T_{2,OBMF}} + \frac{1}{12} \gamma^2 G_p^2 t_{E_p}^2 D_{OBMF} \quad (5.60)$$

where aside from the diffusion coefficient for gas and water, the diffusion coefficients for oil and OBMF are related to their corresponding T_2 relaxation times through the CVM equations.

Furthermore, the constituent viscosity model can be extended to include the live oils by incorporating the empirical correction for gas oil ratio discussed in Chapter 3.

5.7 Logging guidelines

Because of the complexity of NMR logging and the large number of tool parameters, it is essential to have a pre-logging plan to ensure all logging parameters are set properly to obtain the desired information from the logging run. Most NMR logging companies have a pre-logging planning kit which allows the users to calculate the proper logging parameters for the logging run. Here, we shall mention some of the general guidelines prior to the logging job. For those logging parameters which are tool specific, the readers should refer to the instruction manuals supplied by the logging companies.

Knowing the tool characteristics

It is very important to know the tool characteristics, its capabilities, and its limitations, and to determine if the logging objectives can be accomplished using the specific tool. For example, some tools (e.g. Halliburton/Numar's MRIL) are designed as centralized tools, whereas others (e.g. Schlumberger's CMR) are pad-type tools. For the centralized tools, high salinity of the borehole fluid decreases the efficiency of the tool (lowered Quality Factor, Q). The reduced Q of the transmitting/receiving coil results in decreased signal to noise (S/N) ratio. For certain hole sizes, a mud excluder can be slipped over the tool to minimize this problem. In larger holes, it may be necessary to use a fresh water slug for proper operation of the tool.

The pad type tool does not have the borehole salinity problem. However, depending on the depth of investigation, the tool may be probing a shallow depth into the formation, thus measuring the invaded mud filtrate rather than the formation fluid.

Different tools have very different magnetic field distributions. Depending on these magnetic field distributions, some applications of NMR logging discussed previously may be possible, whereas others may not be applicable.

Tool calibrations and preparations

Prior to logging, it is essential to make sure the tool is properly calibrated. Improper calibration can affect the porosity reading. Other preparation procedures are often recommended by the service companies. They need to be closely followed to ensure the quality of the data.

Establishing the proper repeat delay time

The repeat delay time is designated as RD, or sometimes as TR (Time to Repeat), or WT (Wait Time). This is the time between the successive pulse excitations. To ensure that all the spins are fully recovered from the previous excitation and polarized again along the direction of the applied magnetic field, it is essential to have a long enough RD. The value of RD is dictated by the longest T_1 value of the formation. A good rule of thumb is to choose RD to be three times the longest T_1 of the formation fluids. If too short a repeat delay time is used, porosity and free fluid index will be underestimated. If RD is too long, logging speed will be unnecessarily reduced. Some of the modern NMR logging tools have a long magnet that prepolarizes the spins in the formation, thus increases the logging speed.

All the NMR logging service companies have the means to determine the proper RD or wait time from measurements made in the well. These measurements have been adequate in most wells where water based muds were used, and moderate well bore temperatures ($< 200^\circ\text{F}$) were encountered. For higher temperatures and/or oil based muds, it may be necessary to have prior laboratory or field site measurements on reservoir fluids to determine the T_1 and T_2 in order to ensure that the proper repeat delay times are used.

Well site measurements are recommended where a good filtrate sample can be obtained usually from the mud company who uses a filtrate press. Where a good filtrate sample cannot be obtained, mud and produced fluid

samples should be kept and measured later in the laboratory. From these measurements, corrections can later be made to the logs if too short a repeat delay time was used.

Getting a filtrate sample from an oil based mud is difficult but should be available from the mud companies. If a good filtrate sample cannot be obtained from the mud company, it is recommended to centrifuge the samples and draw off the top (less dense) fraction.

Where the surface temperatures are significantly different from those in the wellbore, it is recommended to slowly lower the tool into the hole so as to allow sufficient time for temperature stabilization.

Stationary measurements

Stationary measurements should be performed in the well to obtain an estimate for T_1 and to improve the default setting of the T_1/T_2 ratio. This parameter is often set at 1.5. (NOTE: In some cases, the T_1/T_2 ratio can be much higher than 1.5. This is not a concern if the wait time is sufficient for the logging conditions.) Stationary measurements should be recorded long enough for the signal/noise to reach at least 20.

Log quality controls

The logging engineer (unless told otherwise) will choose a logging speed to minimize acquisition time while maintaining an acceptable resolution and signal to noise ratio. The highest resolution and signal to noise ratio are achieved at the slowest speeds. Consequently, it is recommended (where hole conditions permit) to run the tool as slowly as possible. The data quality improvement can be tested during tool setup by choosing a short interval and running the tool first at the speed recommended by the logging company, and then at a half or a quarter of that speed.

Tool repeatability

The measurement is statistical in nature and does not repeat exactly. When the borehole is not washed out or rugose, porosity should repeat within 1 p.u. and FFI should repeat within 0.5 p.u.

Log data consistency checks

NMR porosity should be comparable to matrix corrected neutron/density porosity in clean zones where there is no heavy oil and gas. Shale and heavy oil will cause a reduction in NMR porosity as compared to crossplot porosity.